

Anand Kumar

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EDUCATION

• Indian Institute of Technology, Bombay (IITB)

November 2020 - April 2024

Bachelor of Technology in Computer Science and Engineering

EXPERIENCE

• Navi Technologies Limited, Bengaluru

July 2024 - Present

Software Development Engineer / UPI Backend

◦ UPI Multibank | Integration of Multiple Payment Service Providers

- * Integrated multiple PSPs into UPI, reducing operational costs by 60% and eliminating single points of failure
- * Built a dedicated PSP communication microservice supporting onboarding and transactions across 30+ APIs, centralizing authentication, retries, and error handling
- * Refactored the SQL schema to support multi-provider UPI flows with configurable routing across PSPs
- * Introduced API-level circuit breakers to isolate downstream failures and prevent cascading outages

◦ Project Nadaga | System Scaling and Optimization for 10% UPI market share

- * Optimized core transaction flow by eliminating unnecessary inter-service hops and caching to reduce repeated downstream calls
- * Improved database connection pool stability by introducing connection-level jitter in timeouts, preventing synchronized connection expiry and resource contention across pods during high traffic
- * Enhanced inter-service communication by migrating to HTTP/2, reducing connection pool usage by 4× and CPU usage by 25%, validated by Gatling load tests and Prometheus metrics

◦ Gratification Service | Automated Recovery for UPI Transactions

- * Designed an event-driven system to track and resolve pending UPI transactions
- * Implemented quota-aware polling to fetch transaction status from NPCI, reducing manual reconciliation
- * Reduced pending transaction resolution time by 10% through automated workflows
- * Improved transaction reliability and customer trust, contributing to a 20% increase in average transaction value

• Nutanix Private Limited, Pune

May 2023 - June 2023

Software Engineering Intern

◦ Snowflake Table Lifecycle Management

- * Delivered a configuration-driven solution to automate identification and truncation of 100+ Snowflake tables
- * Leveraged DynamoDB as a config store to drive dynamic query generation, reducing manual effort by 25%
- * Containerized the service using Docker and deployed it on Kubernetes with ArgoCD, enabling automated, versioned, and controlled rollouts

PROJECTS

• SAT (Boolean Satisfiability) Solver

June 2022 - July 2022

Logic for Computer Science Course

- Built a solver using the Z3 theorem prover in Python to solve k-coloring graphs, Sudoku, and Rush Hour
- Encoded game states into Boolean satisfiability formulas, reducing solver time by 35% through constraint pruning
- Integrated NetworkX to visualize graph-coloring outputs and validate correctness across 200+ test cases

• Peer-to-Peer File Sharing Network

February 2022 - April 2022

Computer Networks Course

- Engineered a P2P file-sharing network using C++ sockets, supporting 20+ concurrent peers
- Applied Dijkstra's algorithm for shortest-path routing of file transfers, improving inter hop latency by 40%
- Used TCP for reliable, ordered transfers with MD5 checksums for transfer integrity verification

TECHNICAL SKILLS

Languages	: Java, Kotlin, Go, Python, C++, SQL, Bash
Backend & Frameworks	: Spring Boot, Kafka, REST APIs, gRPC, Protobuf, Maven
Databases & Storage	: PostgreSQL, Firestore, Redis, DynamoDB, S3, Firebase, Elasticsearch
Infrastructure & Tools	: Docker, Kubernetes, AWS, Git, Linux, Prometheus, Grafana, Loki

ACHIEVEMENTS

- Secured **All India Rank 238** in **JEE Main 2020** and **All India Rank 369** in **JEE Advanced 2020**
- Recipient of the **KVPY Fellowship** (IISc Bangalore) and **NTSE Scholarship** (NCERT)
- Achieved **World Rank 17** in the **International Mathematics Olympiad** conducted by SOF
- Qualified **NSEJS** and **NSEC** Olympiads conducted by IAPT
- Awarded the **Navi Spearhead Award** for high-impact contributions at Navi